STAT 1210-2

Section 2: Probability and Distributions Exercises

Question 1

A network of wildlife cameras operated for 1200 camera-nights. A red fox was recorded during 174 of those nights.

1. Use the observations to estimate the probability that a red fox is recorded during a randomly selected camera-night.
2. Estimate the probability that a red fox is not recorded during a randomly selected camera-night.
3. If similar conditions apply during another 500 camera-nights, how many nights would be expected to contain a red-fox detection?
4. Does the answer to Part 3 guarantee the observed number of detections? Explain using the frequentist interpretation of probability.

Question 2

Two laboratory assays are performed on the same specimen. Each assay is recorded as positive (+) or negative (−), and the order of the assays is retained.

1. Write the sample space for the two assay results.
2. Write the event that exactly one assay is positive.
3. Write the event that at least one assay is positive.
4. Write the complement of the event in Part 3.
5. If each assay has probability 0.20 of a positive result and the assay results are independent, calculate the probability that exactly one assay is positive.

Question 3

Let X be the number of seedlings, out of four, that show measurable root growth after a laboratory treatment. A proposed probability model is shown below.

x	0	1	2	3	4
$P(X = x)$	0.10	0.24	0.32	0.22	c

1. Find c so that the table is a valid probability model.
2. Calculate $P(X \geq 3)$ and $P(X < 2)$.
3. Calculate the mean $\mu_X = E(X) = \sum xP(X = x)$.
4. Calculate the variance and standard deviation of X. Note that you can also calculate variance using this formula

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \sum x^2 P(X = x) - \left(\sum xP(X = x)\right)^2$$

5. Interpret the mean in context. Explain why the mean does not have to be a possible observed count.

Question 4

The calibration error Y of a timing sensor is uniformly distributed between -0.6 and 0.6 microseconds.

1. Find the height of the uniform density curve.
2. Use areas to calculate $P(-0.2 \leq Y \leq 0.3)$.
3. Use areas to calculate $P(Y < -0.45 \text{ or } Y > 0.50)$.
4. Find $P(Y = 0)$ and explain the result.
5. Explain why $P(Y \leq 0.3) = P(Y < 0.3)$.

Question 5

An insurer uses the following simplified probability model for the annual claim cost C of one policy.

Claim cost c	\$0	\$500	\$3000	\$15,000
$P(C = c)$	0.88	0.08	0.03	0.01

1. Calculate the expected annual claim cost $E(C)$.
2. Calculate $E(C^2)$.
3. Use $\text{Var}(C) = E(C^2) - [E(C)]^2$ to calculate the variance and standard deviation.
4. Find the expected total claim cost for 10,000 policies under this model.
5. If the insurer collects a premium of \$350 per policy before administrative expenses, calculate the expected amount remaining per policy after claim costs. Explain why this is a long-run average rather than a guaranteed profit on every policy.

Question 6

Among 2500 animals receiving a veterinary medication, 37 experienced a specified adverse reaction.

1. Calculate the empirical risk of the adverse reaction.
2. Calculate the empirical probability of no adverse reaction.
3. Calculate the odds of an adverse reaction and express them in the form`about 1 to k."
4. Explain the difference between a risk of 0.0148 and odds of approximately $1 : 66.6$.

Question 7

Let X be the number of surviving offspring in a four-offspring litter. The probability model is

x	0	1	2	3	4
$P(X = x)$	0.05	0.15	0.30	0.30	0.20

A rehabilitation program models the litter's cost, in dollars, as

$$Y = 120 + 45X$$

1. Calculate $E(X)$ and $\text{SD}(X)$ (Note: SD here means standard deviation).
2. Calculate $E(Y)$ and $\text{SD}(Y)$ without constructing a new probability table.
3. Calculate $P(Y \geq 255)$.
4. Interpret $E(Y)$ in context.

Question 8

In a population of dogs, let V be the event that a dog is vaccinated and let I be the event that the dog carries a particular infection. Express each statement using set notation.

1. The dog is vaccinated and infected.
2. The dog is vaccinated or infected, including the possibility of both.
3. The dog is vaccinated but not infected.
4. The dog is neither vaccinated nor infected.
5. The dog has at least one of the two characteristics.

Question 9

Researchers examine 400 seabirds. Of these birds, 145 show evidence of plastic ingestion, 92 have low body mass and 55 have both characteristics. Let A denote plastic ingestion and B denote low body mass.

1. Calculate $P(A)$, $P(B)$ and $P(A \cap B)$.
2. Use the general addition rule to calculate $P(A \cup B)$.
3. Calculate the probability that a bird has neither characteristic.
4. Calculate the probability that a bird has plastic ingestion but not low body mass.
5. Explain why simply adding $P(A)$ and $P(B)$ would give an incorrect answer for $P(A \cup B)$.

Question 10

In a population, the probability that an individual carries genetic marker A is 0.08, and the probability of carrying marker B is 0.12. Assume the two marker events are independent.

1. Calculate the probability that an individual carries both markers.
2. Calculate the probability that an individual carries at least one marker.
3. Calculate the probability that an individual carries neither marker.
4. Are the events`carries marker $A$" and`carries marker B" disjoint? Explain.

Question 11

A veterinary hospital records the following results for 300 dogs.

	Diarrhea	No diarrhea	Total
Parasitic infection	48	32	80
No parasitic infection	36	184	220
Total	84	216	300

Let I denote parasitic infection and D denote diarrhea.

1. Calculate $P(I \cap D)$.
2. Calculate and interpret $P(I \mid D)$.
3. Calculate and interpret $P(D \mid I)$.
4. Explain why $P(I \mid D)$ and $P(D \mid I)$ are different even though both use the same intersection count.

Question 12

A wildlife health study records environmental exposure and skin lesions in 300 animals.

	Lesion	No lesion	Total
Exposed	30	70	100
Not exposed	20	180	200
Total	50	250	300

Let E denote exposure and L denote a lesion.

1. Calculate $P(L)$ and $P(L \mid E)$.
2. Based on these probabilities, are E and L independent? Explain.
3. Check independence a second way by comparing $P(E \cap L)$ with $P(E)P(L)$.
4. Does the observed association prove that the exposure caused the lesions? Explain briefly.

Question 13

In a population of plants, 4% carry a pathogen. A screening test has sensitivity 92% and specificity 95%. Consider 10,000 plants representative of this population.

1. Construct the expected counts of true positives, false negatives, false positives and true negatives.
2. Calculate the overall probability of a positive result.
3. Calculate $P(\text{pathogen} \mid +)$ and interpret it.
4. Calculate $P(\text{pathogen} \mid -)$ and interpret it.
5. Explain why high sensitivity and specificity do not necessarily produce a positive predictive value near 100% when prevalence is low.

Question 14

Suppose a DNA profile would match a randomly selected innocent person with probability 1/10,000. A city contains approximately 100,000 people who could not initially be excluded as possible suspects.

1. Among 100,000 innocent people, how many coincidental matches would be expected under this simplified model?
2. Explain the difference between

$$P(\text{match} \mid \text{innocent}) \quad \text{and} \quad P(\text{innocent} \mid \text{match})$$

3. Can the second probability be calculated from the match probability alone? Explain what additional information is needed.
4. State the error made in the prosecutor's fallacy.

Question 15

An animal shelter has 12 cats, of which 8 are vaccinated and 4 are not vaccinated. Three different cats are selected at random without replacement.

1. Calculate the probability that all three selected cats are vaccinated.
2. Calculate the probability that at least one selected cat is not vaccinated.
3. Calculate the probability that exactly two selected cats are vaccinated.
4. Given that the first two selected cats are vaccinated, calculate the probability that the third is vaccinated.
5. Explain why the three selection outcomes are not independent.

Question 16

Resting heart rates in a population of healthy adult horses are approximately Normal with mean 40 beats per minute and standard deviation 6 beats per minute.

1. Find the z-score for a horse with a resting heart rate of 52 beats per minute.
2. Find the z-score for a horse with a resting heart rate of 34 beats per minute.
3. Interpret both z-scores in context.
4. Which observation is more unusual relative to the population distribution? Explain.

Question 17

Let Z have the standard Normal distribution. Use a standard Normal table to calculate each probability.

1. $P(-1.25 < Z < 0.80)$
2. $P(Z > 1.65)$
3. $P(|Z| > 2.00)$
4. Sketch a standard Normal curve and shade the region corresponding to each probability.

Question 18

The fork length X of adult Atlantic salmon returning to a river is approximately Normally distributed with mean 50 cm and standard deviation 6 cm.

1. Calculate $P(X < 44)$.
2. Calculate $P(X > 59)$.
3. Calculate $P(47 < X < 56)$.
4. In a group of 800 returning salmon, approximately how many would be expected to exceed 59 cm?
5. Explain why the expected count in Part 4 need not equal the observed count exactly.

Question 19

An enzyme measurement in a healthy population is approximately Normal with mean 120 units and standard deviation 15 units.

1. Find the 90th percentile of the distribution.
2. Find the value that separates the lowest 5% of measurements from the remaining 95%.
3. Find the two values that contain the central 95% of measurements.
4. Interpret the 90th percentile in context.

Question 20

A dog's fasting glucose measurement is 135 mg/dL. In the relevant healthy dog population, glucose has mean 100 mg/dL and standard deviation 15 mg/dL. A cat's creatinine measurement is 2.1 mg/dL. In the relevant healthy cat population, creatinine has mean 1.4 mg/dL and standard deviation 0.35 mg/dL. Assume both reference distributions are approximately Normal.

1. Calculate the z-score for each measurement.
2. Which measurement is farther above its population mean in standard-deviation units?
3. Calculate the upper-tail probability associated with each value.
4. Explain why the raw measurements cannot be compared directly.

Question 21

The wing span of a bird species is approximately Normal with mean 82 cm and standard deviation 4 cm.

1. Use the 68–95–99.7 rule to give the interval containing approximately 68% of wing spans.
2. Give the interval containing approximately 95% of wing spans.
3. Give the interval containing approximately 99.7% of wing spans.
4. Among 2000 birds, approximately how many would be expected to have wing spans outside the 95% interval?
5. Approximately how many would be expected to have wing spans greater than 90 cm?

Question 22

Recovery time X after a veterinary procedure is approximately Normal with mean 50 minutes and standard deviation 8 minutes.

1. Calculate $P(X > 60)$.
2. Calculate $P(X > 45)$.
3. Among animals whose recovery time exceeds 45 minutes, calculate the probability that recovery exceeds 60 minutes.
4. Find the recovery time that marks the slowest 10% of recoveries.
5. Explain the difference between the unconditional probability in Part 1 and the conditional probability in Part 3.

Question 23

For each setting, state whether the count is reasonably modelled as binomial, Poisson or neither. Give a brief reason.

1. The number of 20 independently treated patients who experience an adverse reaction, when each patient has the same reaction probability.
2. The number of emergency calls received during one hour, assuming a stable average rate and independent occurrences.
3. The number of tagged fish in a sample of 10 fish drawn without replacement from a small tank containing 15 tagged and 15 untagged fish.
4. The number of patients treated until the first successful response occurs.
5. The number of bacterial colonies appearing on a fixed-area plate, assuming colonies arise independently at a stable average rate across the plate.

Question 24

Each of 12 independently treated animals has probability 0.80 of producing a protective immune response. Let X be the number producing a protective response.

1. State the probability distribution of X.
2. Calculate $P(X = 10)$ using the binomial formula.
3. Calculate the probability that all 12 animals produce a protective response.
4. Find the mean and standard deviation of X.
5. Interpret the mean in context.

Question 25

A manufacturer reports that each diagnostic cartridge has probability 0.04 of being defective. Assume cartridge outcomes are independent. A laboratory receives 20 cartridges, and X is the number that are defective.

1. State the binomial model for X.
2. Calculate $P(X = 0)$ and $P(X = 1)$.
3. Calculate $P(X \geq 2)$ using a complement.
4. Find the mean and standard deviation of X.
5. Explain why the mean does not imply that a box can contain 0.8 of a defective cartridge.

Question 26

The number X of policyholders making a specified type of claim in a group of 50 independent policyholders is modelled as binomial. The expected number of claims is 5.

1. Use the binomial mean to find the claim probability p.
2. Find the standard deviation of X.
3. Calculate the probability of no claims.
4. Calculate the probability of at most two claims.
5. Explain what assumption would be questionable if a single storm could cause many of the policyholders to claim at the same time.

Question 27

Wildlife–vehicle collisions on a rural highway are modelled by a Poisson process with an average of 3.6 collisions per week. Let X be the number of collisions during a two-day period.

1. Find the appropriate value of λ for two days.
2. Calculate $P(X = 0)$.
3. Calculate $P(X = 1)$.
4. Calculate $P(X \geq 2)$.
5. State the mean and standard deviation of the two-day count.

Question 28

Under a baseline model, the monthly number X of seabird deaths found along a monitored coastline follows a Poisson distribution with $\lambda = 2$.

1. Calculate $P(X \leq 1)$.
2. Calculate $P(X \geq 7)$.
3. Find the smallest whole number k for which $P(X \geq k) < 0.01$.
4. If seven deaths are observed, explain what the small probability does and does not establish.

Question 29

The number of ticks collected along a 100-metre forest transect is modelled as Poisson with mean 1.5. Assume the same rate applies along a comparable 300-metre transect.

1. Find the Poisson parameter for the 300-metre transect.
2. Calculate the probability of exactly five ticks along 300 metres.
3. Calculate the probability of at least eight ticks along 300 metres.
4. Find the mean, variance and standard deviation of the 300-metre count.
5. Identify one real-world feature that could make the Poisson model inappropriate.

Question 30

In a population of animals, 12% carry an infection. A screening test has sensitivity 90% and specificity 95%. Fifteen animals are independently selected from the population and tested. Let Y be the number of positive test results.

1. Use the law of total probability to find the probability that one randomly selected animal tests positive.
2. State an appropriate probability model for Y.
3. Calculate $P(Y \geq 3)$.
4. Find the mean and standard deviation of Y.
5. Given that one randomly selected animal tests positive, calculate the probability that it actually carries the infection.
6. Explain why the probability in Part 5 differs from the sensitivity.

Question 31

A marine research team tags Atlantic salmon before migration. Let R be the event that a tagged salmon returns to the river the following year, and let T be the event that its transmitter is still functioning when it returns. Suppose

$$P(R) = 0.62, \qquad P(T) = 0.80, \qquad P(R \cap T) = 0.54$$

1. Calculate $P(R \cup T)$.
2. Calculate $P(R \mid T)$ and interpret the result in context.
3. Calculate the probability that neither event occurs.
4. Are R and T disjoint? Explain.
5. Are R and T independent? Support your answer with a calculation.

Question 32

Let X be the number of emergency patients arriving at a veterinary clinic during a randomly selected two-hour period. The proposed probability model is shown below.

x	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	c

1. Find c so that this is a valid probability model.
2. Calculate $P(X \geq 2)$ and $P(1 \leq X < 4)$.
3. Calculate the mean $E(X)$.
4. Calculate the variance and standard deviation of X.
5. Interpret $E(X)$ in the context of many two-hour periods.

Question 33

A current meter used in an oceanography experiment has a measurement error Y, in centimetres per second, that is uniformly distributed between -0.3 and 0.5.

1. Find the height of the uniform density curve.
2. Calculate $P(-0.1 \leq Y \leq 0.2)$ using the area under the density curve.
3. Calculate $P(Y < -0.2 \text{ or } Y > 0.35)$.
4. Calculate $P(Y < 0.1 \mid Y > -0.1)$.
5. What is $P(Y = 0.1)$? Explain using the geometry of a continuous probability distribution.

Question 34

An insurance company uses the following simplified model for a pet owner's eligible veterinary expenses C during one year.

Eligible expenses c	\$0	\$400	\$1200	\$4000
$P(C = c)$	0.55	0.25	0.15	0.05

The policy has a \$500 annual deductible and pays all eligible expenses above the deductible. The insurer's payment is therefore $Y = \max(C - 500, 0)$.

1. Construct the probability distribution of Y.
2. Calculate the expected annual eligible expense $E(C)$.
3. Calculate the insurer's expected annual payment $E(Y)$.
4. Calculate the owner's expected annual out-of-pocket amount $E(C - Y)$.
5. Explain why the expected payment is not the amount the insurer will pay for every policy.

Question 35

A wildlife health study records whether each of 500 seabirds had substantial exposure to an environmental contaminant and whether a particular liver lesion was present.

	Lesion present	Lesion absent	Total
Substantial exposure	72	108	180
No substantial exposure	48	272	320
Total	120	380	500

Let E denote substantial exposure and let L denote the presence of the lesion.

1. Calculate $P(L \mid E)$ and $P(L \mid E^c)$.
2. Calculate the risk ratio $P(L \mid E)/P(L \mid E^c)$ and interpret it.
3. Calculate $P(E \mid L)$.
4. Are E and L independent? Support your answer numerically.
5. Can this observational study establish that the contaminant caused the lesion? Explain one limitation.

Question 36

In a population of cats, 8% have early kidney disease. A screening test has sensitivity 91% and specificity 94%.

1. For a hypothetical population of 10,000 cats, find the expected numbers of true positives, false negatives, false positives and true negatives.
2. Calculate the probability that a randomly selected cat tests positive.
3. Given a positive result, calculate the probability that the cat has early kidney disease.
4. Given a negative result, calculate the probability that the cat does not have early kidney disease.
5. Explain why false positives may outnumber false negatives even though the test has high sensitivity and specificity.

Question 37

Daily milk production for cows in a particular herd is approximately Normally distributed with mean 31.5 litres and standard deviation 4.2 litres.

1. Calculate the probability that a randomly selected cow produces between 28 and 36 litres in one day.
2. Find the daily production level that separates the highest 10% of cows from the remaining 90%.
3. Among 800 cows, approximately how many would be expected to produce less than 24 litres in one day?
4. If 2.5 litres are added to every recorded value because of a calibration correction, state the new mean and standard deviation.
5. Explain why the calculation in Part 4 changes the mean but not the standard deviation.

Question 38

A wildlife vaccination program distributes oral vaccine baits. Under similar field conditions, each of 14 independently monitored raccoons has probability 0.72 of developing the intended immune response. Let X be the number that develop the response.

1. State the probability distribution of X and identify its parameters.
2. Calculate $P(X = 10)$.
3. Calculate $P(X \geq 11)$.
4. Calculate $P(8 \leq X \leq 12)$.
5. Find the mean and standard deviation of X and interpret the mean.
6. Identify two assumptions required for the binomial model to be appropriate.

Question 39

An insurance claims office receives an average of 1.8 new claims during an eight-hour workday. Assume that claims arrive according to a Poisson process at a constant rate.

1. Find the Poisson parameter for a three-hour period.
2. Calculate the probability of receiving no claims during a three-hour period.
3. Calculate the probability of receiving at least two claims during a three-hour period.
4. Find the Poisson parameter for five complete workdays and calculate the probability of receiving between 6 and 12 claims, inclusive.
5. Identify one realistic feature of claim arrivals that could violate the constant-rate assumption.

Question 40

A wildlife disease program monitors animals from two regions. Region A contains 40% of the monitored population and has infection prevalence 18%. Region B contains the remaining 60% and has infection prevalence 6%. A diagnostic test has sensitivity 92% and specificity 96% in both regions. Let A denote selection from Region A, let I denote infection and let $+$ denote a positive test result.

1. Use the law of total probability to calculate $P(I)$.
2. Calculate $P(+ \mid A)$ and $P(+ \mid A^c)$.
3. Calculate the overall probability $P(+)$.
4. Given a positive result, calculate $P(A \mid +)$.
5. Given a positive result, calculate $P(I \mid +)$.
6. If 20 animals are independently selected from this mixed population and tested, state a probability model for the number of positive results and calculate the probability of at least three positive results.
7. Find the expected number of positive results among the 20 tests.